Aerial Image Object Detection Based on RGB–Infrared Multibranch Progressive Fusion

Kewei Liu, Tao Li, and Dongliang Peng

***Abstract*— In RGB–infrared aerial image object detection, fully utilizing the advantages of both RGB and infrared images for effective detection is a key challenge in this field. In response to the challenges outlined, an RGB–infrared multibranch progressive feature fusion method for object detection in aerial images is proposed. Specifically, since accurate detection of objects in aerial images usually requires extensive context information, considering the connection between objects and backgrounds, the global–local synergistic attention (GLSA) is constructed. To mitigate the impact of feature domain differences between RGB and infrared images during the complementary information filtering process, a multimodal complementary information filter (MCIF) for RGB and infrared images is designed on the basis of GLSA. To fully exploit the advantages of the two types of model images, a multibranch progressive feature fusion network is designed. The multibranch progressive feature fusion network progressively fuses the enhanced features from the GLSA with the filtered complementary information by the MCIF, resulting in a final fused feature that integrates the rich visual details of RGB images with the distinctive physical characteristics of infrared images. Results against publicly released datasets demonstrate that the proposed method achieves state-of-the-art detection performance.**

***Index Terms*— Aerial images, complementary fusion, multispectral object detection, RGB–infrared object detection.**

I. Introduction

AERIAL image object detection is one of the classic tasks in the field of remote sensing, primarily aiming at identifying the location and category of objects of interest in aerial images. This technology has gained widespread application in various domains such as smart cities, intelligent security, rescue operations, and disaster relief [1].

Aerial image object detection has traditionally been centered on single-modal imaging, particularly visible and infrared modalities [2], [3]. Visible images, renowned for their rich texture and high spatial resolution, face significant challenges under complex environmental conditions, such as low illumination, haze, or occlusions. Infrared images, on the other hand, excel in detecting objects under low-light conditions due to their reliance on thermal signatures [4]. However, their lower resolution and lack of color and texture information make them less effective for detecting small, low-contrast objects and scenes with intricate structural details. These limitations hinder their reliability in real-world applications [5], [6], [7], [8].

Research has extensively explored methodologies to address these constraints. Techniques like S2ANet [9], DRN [10], SCRDet [11], OrientedRepPoints [12], and RLPGBNet [13] focus on improving feature alignment, integrating attention mechanisms, and enhancing representation capabilities for visible imagery, achieving notable successes in conventional scenarios. However, their efficacy diminishes in challenging conditions, such as low-light environments and cluttered backgrounds, where single-modal imagery often falls short [14].

Recent efforts have sought to overcome these inherent limitations by leveraging multimodal data fusion. By integrating complementary information from texture-rich visible images and infrared images that are robust to environmental variations, researchers strive to substantially improve object detection accuracy across diverse scenes and conditions. The FCRPN method achieves better object detection than single-modal images by fusing RPN information from infrared and visible images [15]. The MLF-FRCNN method constructs a multilayer fusion network that creates a multiscale feature map based on the infrared and visible feature information acquired by the backbone, improving detection in low-light environments [16]. The CMIA method introduces a cross-modal interactive attention module that dynamically modifies channel weights by establishing correlations between distinct modal feature channels to maximize the utilization of multimodal information [17]. A feature-enhanced remote attention fusion network is proposed in the LRAF-Net method, which improves the detection performance by fusing the remote dependence of enhanced visible and infrared features [18]. Studies such as FCRPN and MLF-FRCNN underline the potential of such multimodal approaches, illustrating promising advancements in overcoming single-modal deficiencies.

Although different modal images provide complementary information, a significant amount of redundant information exists between modalities. This redundancy can diminish the importance of the unique information contributed by each modality. Existing multimodal image fusion object detection methods struggle to accurately filter complementary

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information, making it difficult to mitigate the negative effects of redundant information. Furthermore, methods such as FCRPN, MLF-FRCNN, and CMIA primarily focus on multimodal image fusion for object detection in natural images. Compared to natural images, aerial images present unique challenges, including smaller object sizes and more complex backgrounds, making the direct application of multimodal fusion detection algorithms developed for natural images to aerial images less effective. These challenges significantly increase the difficulty of achieving high-accuracy object detection in aerial imagery.

To overcome the aforementioned challenges, we propose an aerial image object detection method based on multibranch progressive feature fusion (ADMPF) of infrared and RGB images. In this method, we propose a global–local synergistic attention (GLSA) mechanism that models the relationship between the background and the object. Building on this, we develop a multimodal complementary information filter (MCIF) using conditional probability. The GLSA enhances the single-modal features of RGB and infrared images, enabling the model to focus more effectively on key object areas. The MCIF selectively extracts crucial complementary information between the features of RGB and infrared images. Subsequently, we design a multibranch progressive feature network (MPFN) to integrate the outputs of the GLSA and MCIF. This network employs different branches for the initial fusion of features enhanced by the GLSA and the results filtered by the MCIF, followed by a deeper fusion of these preliminary fusion results. This process ensures that the final fused feature effectively combines the rich visual details of RGB images with the unique physical characteristics of infrared images.

The main contributions of this article are summarized below.

1) In view of the characteristics of the object distribution in aerial images, a GLSA is proposed. This attention combines both global and local contextual relationships, enabling it to not only capture key information relevant to the object detection task but also serve as a foundation for filtering the complementary information of multimodal image features.
2) Given that multimodal images contain both complementary and redundant information, an MCIF is proposed. By selectively filtering the crucial complementary information between modalities, it establishes a foundation for the effective fusion of multimodal images.
3) Based on the principle of progressive fusion, a multibranch fusion network shaped like a binary tree is designed for the fusion of RGB–infrared image features. This fusion structure decouples the fusion paths of comprehensive information and key complementary information and then integrates the fused comprehensive and complementary information, thus achieving the efficient extraction and utilization of feature information from comprehensive to key aspects.

The rest of the article is organized as follows. Section II reviews related works on multimodal complementary information applications and attention mechanisms. Section III details the specific components of the proposed ADMPF method such as complementary information filters and fusion networks. Subsequently, Section IV presents empirical results to confirm the efficacy of the proposed ADMPF. Finally, Section V concludes the ADMPF method.

II. Related Work

The information contained in images of different modalities is different, including pixel intensity, contrast, texture, and so on. Making full use of complementary information between images of different modalities is an important way to obtain comprehensive information about objects. Section II describes the research related to the localization of complementary and key information in multimodal image fusion.

A. RGB–Infrared Fusion for Aerial Image Object Detection

RGB–infrared fusion-based aerial image object detection leverages the complementary advantages of visible and infrared images, demonstrating significant potential in detecting objects in complex scenarios. To address the critical challenges in multimodal feature fusion, various innovative approaches have been proposed. Zhang et al. [19] developed SuperYOLO by integrating a super-resolution branch into the object detection model, enhancing the representation and detection accuracy of small objects through the fusion of infrared and visible images. Sun et al. [20] proposed UA-CMDet that incorporates an uncertainty-aware module to quantify detection confidence across different modalities, effectively reducing the impact of low-confidence objects on multimodal detection performance. Yuan and Wei [21] introduced C^2Former, a Transformer framework with an intermodality cross-attention module and an adaptive feature sampling module. This framework efficiently calibrates modal features, extracts complementary information, and reduces computational costs. Wang et al. [22] designed FFODNet that combines a joint expression optimization module and a specific information enhancement module to suppress interference from modal features and enhance discriminative object features, thereby improving the quality of multimodal fusion.

In contrast to these methods, the approach proposed in this article introduces a novel strategy for efficiently utilizing complementary and redundant information. By accurately filtering and independently fusing these features, the proposed method improves feature utilization efficiency and significantly enhances object detection performance.

B. Complementary Information Enhanced Fusion

Influenced by the current differential amplifier, Zhou et al. [23] proposed a differential modality-aware fusion (DMAF) module, which uses differential modal information to enhance different modalities. Specifically, the DMAF suppresses the common information between different modalities while amplifying the differential information. The common information partially reflects the mutual features of the different modalities. In contrast, the differential information represents the unique features captured by each modality. The enhancement of differential information increases the sensitivity of

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the network to the feature information extracted from another modality. Qingyun and Zhaokui [24] proposed a cross-modal attention feature fusion module to take advantage of the complementarity between different modalities. The module consists of a differential enhancement module and a public selection module. The differential enhancement module discerns differences between modalities to form differential modalities, based on which the channel attention can be acquired to act upon the original features to boost disparity information. Conversely, by summing distinct modality features, the public selection module computes the common modality, with which a channel attention mechanism is calculated to filter out effective information.

Due to differences between the spatial feature distribution of infrared and visible images, the reliability of methods that obtain complementary information by subtraction between the features of infrared and that of visible images is relatively low. Therefore, the above fusion methods based on differential operation may not acquire satisfactory complementary information from different modalities.

C. Attention Mechanism

The attention mechanism has been designed to filter out noise from complex feature information and amplify pertinent information relevant to the current task.

Woo et al. [25] proposed a hybrid attention mechanism called CBAM, which combines channel attention with spatial attention. The CBAM enhances the feature representation of different channels while extracting key information at different spatial locations, thus aggregating crucial feature information from both the channel and space. Wang et al. [26] proposed EcaNet, a channel attention mechanism for local cross-channel information interaction, which effectively improves the performance of various deep CNN frameworks. The channel weights of EcaNet are computed from the 1-D convolution at channel dimensions. To be specific, the kernel size of the 1-D convolution is adaptively determined according to the nonlinear set of the input feature channel dimension, and the coverage of local cross-channel interaction is determined by the convolution kernel size. Hou et al. [27] proposed a coordinate attention (CA) mechanism, where the input features are aggregated into vertically and horizontally oriented attention maps using two 1-D global pooling operations. Therefore, the generated attention maps allow the network to capture long-range correlations along the horizontal direction while preserving location information against the corresponding vertical direction. The pooling of the two directional attention maps emphasizes the information of interest in the feature maps.

In this article, to extract key information from single-modal aerial images, we have developed a GLSA mechanism. The GLSA mechanism works by establishing links between feature contexts based on the characteristics of aerial images.

III. Proposed Method

The method proposed in this article aims to improve the detection accuracy and robustness of ground objects in aerial images by leveraging multimodal aerial image information (Fig. 1). Specifically, a Siamese backbone network is employed to extract multiscale features from RGB and infrared images. The Siamese network is a model composed of two branches with identical structures and shared weights [28]. In this study, the Siamese network's branches, respectively, receive visible images and infrared images, extracting features using identical neural network structures and parameters. By employing a shared-parameter architecture, the Siamese network not only reduces the number of training parameters but also enhances the generalizability of the features extracted by the different branches. The GLSA mechanism is then designed to combine global and local contextual information, enhancing the association between the object and its background. Building on the GLSA, an MCIF is designed to selectively filter the multiscale features of each modality, extracting crucial complementary information at different scales. Subsequently, a multibranch progressive fusion network (MPFN) is proposed. The MPFN uses different branches to fuse the different modal features enhanced by the GLSA and the crucial complementary information of different modalities obtained by filtering through the MCIF, respectively. The fusion results from the different branches are then further integrated. Finally, the fused multiscale features are refined using PaFPN [29] and fed into the detection head to produce the final detection result.

A. Global–Local Synergistic Attention

Aerial images typically cover large geographic areas, contain objects of various scales, and present complex backgrounds. The GLSA mechanism is designed to simultaneously focus on both global information and local details (Fig. 2). The global context provides a broad perspective, capturing extensive contextual information that helps the model understand the relationship between the object and its background. Meanwhile, the local context hones in on detailed regions, enabling the model to detect small objects and subtle features, which reduces background noise interference and improves detection accuracy. Unlike previous attention mechanisms, the GLSA proposed in this article aggregates global and local information across the three dimensions of width, height, and channel for a more comprehensive feature representation.

For the feature F of a single-modal image, the number of channels is C, the height is H, and the width is W. In the GLSA mechanism, the feature F is first aggregated along the height and width dimensions to obtain the aggregated feature vectors F_h^{avg} and F_w^{avg}. F_h^{avg} and F_w^{avg} compress the information across the entire feature map in the H and W dimensions, respectively, producing representative values that help the model capture the global contextual relationships. This is critical for understanding the overall characteristics of the scene. Since F_h^{avg} focuses on the overall statistical characteristics along the height H direction, and F_w^{avg} focuses on the width W direction, the model becomes more robust to subtle variations in the image background

$$F_h^{\text{avg}} = h_\text{avg}(F) \tag{1}$$

$$F_w^{\text{avg}} = w_\text{avg}(F). \tag{2}$$

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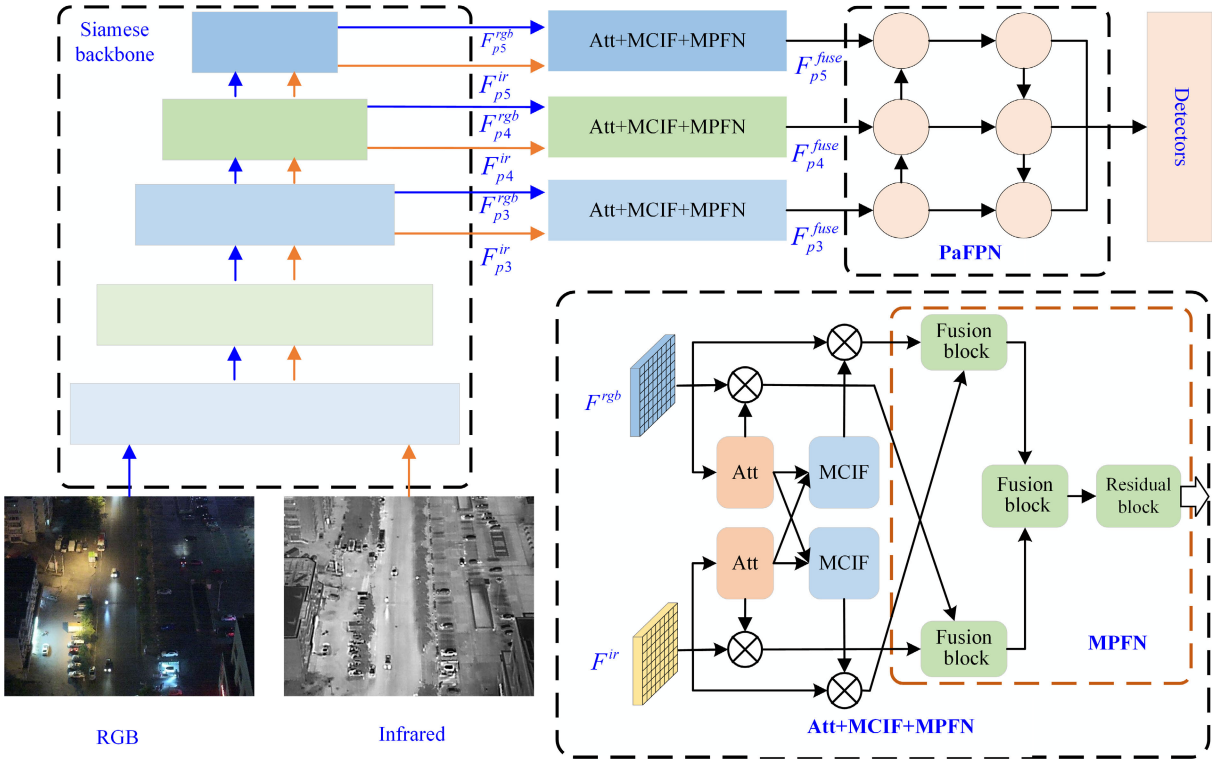


Fig. 1. Framework diagram of the ADMPF. "RGB" and "infrared" in the figure represent the input visible and infrared images. "GLSA" refers to the global–local synergistic attention, "MCIF" stands for the multimodal complementary information filter, and "MPFN" denotes the multibranch progressive fusion network. "Siamese backbone" is a dual-branch network structure with shared weights, where modules in different colors indicate features of varying depths and scales. "PaFPN" is a multiscale feature fusion network that integrates top-down and bottom-up paths with lateral connections. F_{p5}^{rgb}-F_{p3}^{rgb}, F_{p5}^{ir}-F_{p3}^{ir}, and F_{p5}^{fuse}-F_{p3}^{fuse} represent multiscale RGB features, infrared features, and RGB–infrared fusion features, respectively. $\otimes$ represents multiplication.

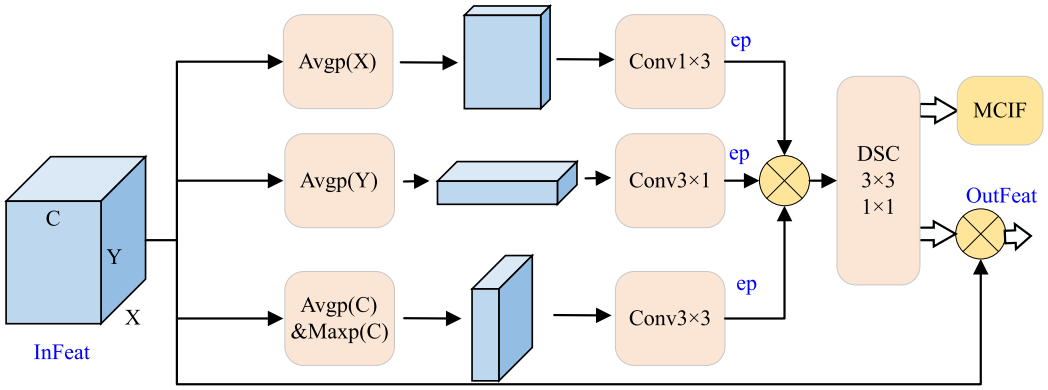


Fig. 2. Framework diagram of the GLSA. InFeat is the input single-modal image feature and OutFeat is the feature after the action of GLSA. Avgp() is the average pooling function. Maxp() is the maximum pooling function. "ep" is the broadcasting mechanism function. DSC is the depth separable convolution function.

Among them, the h_avg$(\cdot)$ and w_avg$(\cdot)$ functions represent the average pooling along the height and width directions, respectively.

Based on F_h^{avg} and F_w^{avg}, small convolutional kernels of 3×1 and 1×3 are applied to process the feature maps along the height and width dimensions, respectively. This allows the model to capture local details while preserving the global contextual relationships

$$G_h = \text{Con3} \times 1\left(F_h^{\text{avg}}\right) \tag{3}$$

$$G_w = \text{Con1} \times 3\left(F_w^{\text{avg}}\right) \tag{4}$$

where G_h and G_w are intermediate features generated during the computation of the GLSA mechanism. Since F_h^{avg} captures long-range contextual relationships along the height (H) direction, G_h emphasizes locally important information in the width (W) direction while preserving the context relationships in the H direction. Similarly, G_w enhances locally important information in the height (H) direction while maintaining the contextual relationships along the W direction.

Although G_h and G_w establish both long-range and local dependencies, they weaken the spatial positional information. To strengthen the position information, the channel information of feature F can be aggregated and then fused with G_h

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and G_w to obtain the final attention map

$$\text{Att} = \text{DSC}(\text{ep}(G_w) \times \text{ep}(G_h) \times \text{ep}(S)) \tag{5}$$

where Att integrates local and remote dependencies while preserving the spatial location relationships between features. Here, S denotes the aggregated channel information for feature F. DSC represents the depth separable convolution with kernel sizes of 3×3 and 1×1. The broadcasting mechanism ep plays a role in carrying out dimensional expansion

$$S = \text{Conv3} \times 3\big(\text{Concat}\big(F_c^{\max}, F_c^{\text{avg}}\big)\big) \tag{6}$$

where $F_c^{\max}$ and F_c^{avg} denote the maximum and the average pooling operations of feature F along the channel dimension, respectively. Besides, Concat conducts the channel concatenation. Average pooling captures the overall statistical characteristics across channels, reflecting the average response intensity of channel features, while max-pooling emphasizes the most prominent features. Combining these two pooling methods enhances the model's robustness to noise and minor variations in channel features. The prior pooling operation models the long-range contextual relationships in the channel direction, while the 3×3 convolution further models local spatial relationships in the feature map, building upon the established context.

The GLSA effectively merges global and local information through cross-dimensional interaction and fusion, enhancing the model's focus on key information and improving feature representation. Additionally, it lays a foundation for the subsequent complementary information filter.

B. Multimodal Complementary Information Filter

Images from different modalities exhibit significant differences in imaging mechanisms and information representation. Previous methods primarily compute complementary information between different modal images based on simple numerical differences, which overlook the complex correlations between infrared and visible images. Given the substantial variations in feature space across modalities, simple difference-based calculations often fail to fully capture the deep semantic and structural information shared between images. Moreover, the difference method typically operates in a local, noncontextual manner, neglecting the relationship between local and global features. As a result, it may not fully leverage the complementary information available in the global context of multimodal images.

The MCIF, built upon the GLSA mechanism, addresses these issues. This attention mechanism simultaneously focuses on both the overall structure and local details of the image, automatically learning the significance of different regions and highlighting key information. By using the GLSA to calculate complementary information, the semantic differences between infrared and RGB images are accentuated. These semantic differences provide rich complementary information, enabling more efficient utilization of critical object-related information within the image (Fig. 3).

The unique information from different modal images constitutes the complement of information. This complementary information includes both crucial object-related data and irrelevant, nonobject data. Fully exploiting the key object-related information in the complementary set is essential to improving cross-modal object detection performance. Specifically, if there exists key information at the same spatial location of infrared images and RGB images, there is information redundancy in that spatial location. However, for the same spatial location, if there is key information in the infrared image feature and invalid information in the RGB image feature, it means the existence of complementary information for the RGB image feature at that spatial location of the infrared image feature and vice versa.

The spatial dimension distribution of key information Att is calculated by the GLSA mechanism. Ranging in $[0-1]$, $\text{Att}_{i,j,z}$ reflects the importance of the feature information in the ith row, jth column, and zth channel in the spatial dimension. Therefore, for the ith row, jth column, and zth channel of paired feature maps of the RGB modality, the corresponding complementarity weight of the infrared image feature can be denoted as

$$\begin{aligned} Pc_{i,j,z}^{\text{ir}} &= \text{Tanh}\big(\lambda \times \text{Att}_{i,j,z}^{\text{ir}} \times \big(1 - \text{Att}_{i,j,z}^{\text{vis}}\big)\big) \\ &= \frac{e^{\lambda \times \text{Att}_{i,j,z}^{\text{ir}} \times \left(1-\text{Att}_{i,j,z}^{\text{vis}}\right)} - e^{-\lambda \times \text{Att}_{i,j,z}^{\text{ir}} \times \left(1-\text{Att}_{i,j,z}^{\text{vis}}\right)}}{e^{\lambda \times \text{Att}_{i,j,z}^{\text{ir}} \times \left(1-\text{Att}_{i,j,z}^{\text{vis}}\right)} + e^{-\lambda \times \text{Att}_{i,j,z}^{\text{ir}} \times \left(1-\text{Att}_{i,j,z}^{\text{vis}}\right)}}. \end{aligned} \tag{7}$$

Among them, Tanh() is the hyperbolic tangent function, and its role is to normalize the complementary weight. The parameter λ is a scale adjustment factor, used to amplify the differences in complementary weights, and is set to 2 in this article.

Similarly, based on the importance of feature information contained in the related infrared image feature, the complementarity weight of the visible image feature stored in the zth channel of the jth column of the ith row is computed as follows:

$$\begin{aligned} Pc_{i,j,z}^{\text{vis}} &= \text{Tanh}\big(\lambda \times \text{Att}_{i,j,z}^{\text{vis}} \times \big(1 - \text{Att}_{i,j,z}^{\text{ir}}\big)\big) \\ &= \frac{e^{\lambda \times \text{Att}_{i,j,z}^{\text{vis}} \times \left(1-\text{Att}_{i,j,z}^{\text{ir}}\right)} - e^{-\lambda \times \text{Att}_{i,j,z}^{\text{vis}} \times \left(1-\text{Att}_{i,j,z}^{\text{ir}}\right)}}{e^{\lambda \times \text{Att}_{i,j,z}^{\text{vis}} \times \left(1-\text{Att}_{i,j,z}^{\text{ir}}\right)} + e^{-\lambda \times \text{Att}_{i,j,z}^{\text{vis}} \times \left(1-\text{Att}_{i,j,z}^{\text{ir}}\right)}}. \end{aligned} \tag{8}$$

C. Multibranch Progressive Feature Network

The MPFN is designed to maximize information retention while minimizing redundancy. The complete information set, consisting of all the information from different modal images, provides a comprehensive view that supports more accurate object detection. However, the intersection of information by different modalities often contains significant redundancy. For the object detection task, this redundant information diminishes the importance of the unique modal-specific information, complicating the model optimization process.

The MPFN proposed in this article is based on a tree-like structure and fuses the features of infrared and visible images at the same scale. The network consists of two branches: a normal fusion branch and an enhanced fusion branch. The normal fusion branch fuses the infrared and RGB image features after applying the GLSA mechanism, while the enhanced fusion branch fuses the crucial complementary information from these images after being filtered by the complementary

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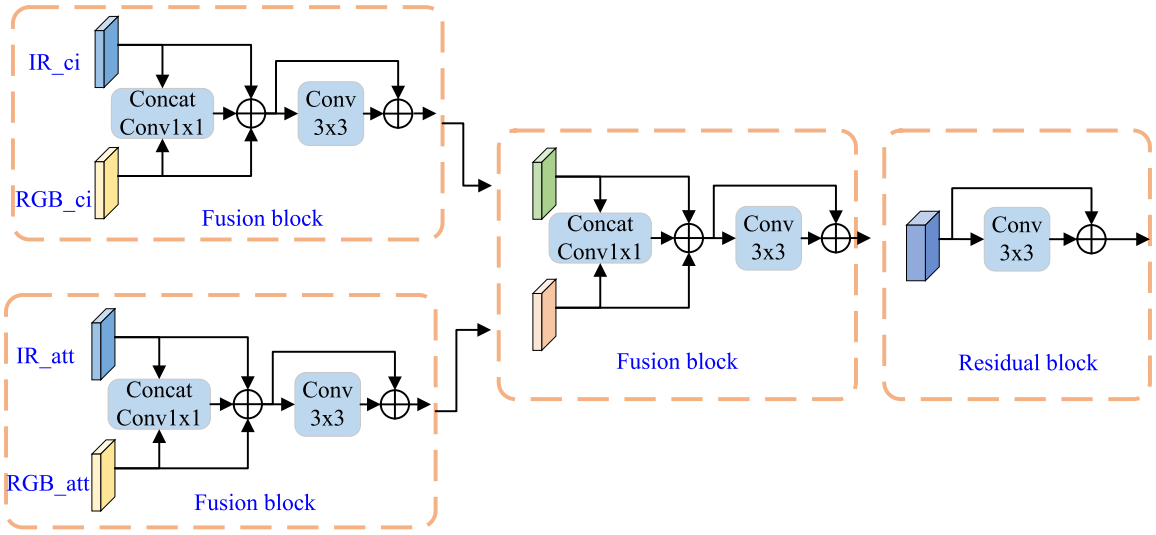


Fig. 3. Framework of the MPFN. "IR_ci" represents the complementary information of infrared image features relative to RGB image features and "RGB_ci" represents the complementary information of RGB image features relative to infrared image features. "IR_att" refers to the infrared image feature after being processed by the GLSA, while "RGB_att" refers to the RGB image feature after the action of the GLSA.

information filter. The attention mechanism highlights key information, reduces the interference of irrelevant details, and improves the quality of the fusion process. Meanwhile, the complementary information filter ensures that unique features from different modalities are preserved and not lost during fusion.

The multibranch design enables the model to fuse features from different perspectives, facilitating the capture of richer and more detailed information. Both the normal and enhanced fusion branches are structured as binary trees, and their final fusion process also follows a binary tree structure. As a result, the entire MPFN forms a tree-like architecture. The normal fusion branch is responsible for obtaining the complete set of information from both RGB and infrared images, providing comprehensive data for the object detection task. However, this complete set also includes an intersection of information, leading to a significant amount of redundancy. The enhanced fusion branch addresses this by emphasizing the complementary information from RGB and infrared images, thereby promoting the efficient use of key information across different modalities while reducing the impact of redundant information.

In the normal fusion branch, the input infrared and visible image features are fused by the convolutional layer of the residual structure after the action of the GLSA mechanism. The fusion result of the normal fusion branch is F^{fn}, which can be computed as

$$F^{\mathrm{fn}'} = \mathrm{Conv}^{1\times1}\left(\mathrm{Cat}\left(F^{\mathrm{att_ir}}, F^{\mathrm{att_vis}}\right)\right) + F^{\mathrm{att_ir}} + F^{\mathrm{att_vis}} \tag{9}$$

$$F^{\mathrm{fn}} = \mathrm{Conv}^{3\times3}\left(F^{\mathrm{fn}'}\right) + F^{\mathrm{fn}'} \tag{10}$$

where $F^{\mathrm{fn}'}$ represents the preliminary fusion result based on the dual-residual structure. $F^{\mathrm{att_ir}}$ and $F^{\mathrm{att_vis}}$ represent the infrared and RGB image features after applying the GLSA mechanism, respectively,

$$F^{\mathrm{att_ir}} = F^{\mathrm{ir}} \times \mathrm{Att}^{\mathrm{ir}} \tag{11}$$

$$F^{\mathrm{att_vis}} = F^{\mathrm{vis}} \times \mathrm{Att}^{\mathrm{vis}} \tag{12}$$

where F^{ir} and F^{vis} denote the infrared image feature and visible image feature, respectively.

In the enhancement fusion branch, the input infrared and RGB image features are fused using the convolutional layer with the residual structure after a key complementary information filter. The fusion result of the enhancement fusion branch is F^{fc}, which is calculated as follows:

$$F^{\mathrm{fc}'} = \mathrm{Conv}^{1\times1}\left(\mathrm{Cat}\left(F^{\mathrm{fc_ir}}, F^{\mathrm{fc_vis}}\right)\right) + F^{\mathrm{fc_ir}} + F^{\mathrm{fc_vis}} \tag{13}$$

$$F^{\mathrm{fc}} = \mathrm{Conv}^{3\times3}\left(F^{\mathrm{fc}'}\right) + F^{\mathrm{fc}'} \tag{14}$$

where $F^{\mathrm{fc}'}$ represents the preliminary fusion result based on the dual-residual structure. $F^{\mathrm{fc_ir}}$ and $F^{\mathrm{fc_vis}}$ are the infrared and visible image features after the action of the key complementary information filters, respectively,

$$F^{\mathrm{fc_ir}} = F^{\mathrm{ir}} \times Pc^{\mathrm{ir}} \tag{15}$$

$$F^{\mathrm{fc_vis}} = F^{\mathrm{vis}} \times Pc^{\mathrm{vis}}. \tag{16}$$

After the normal fusion branch and the enhanced fusion branch are fused, the fusion results from both branches are combined again

$$F^{f_\mathrm{out}'} = \mathrm{Conv}^{1\times1}\left(\mathrm{Cat}\left(F^{\mathrm{fn}}, F^{\mathrm{fc}}\right)\right) + F^{\mathrm{fn}} + F^{\mathrm{fc}} \tag{17}$$

$$F^{f_\mathrm{out}''} = \mathrm{Conv}^{3\times3}\left(F^{f_\mathrm{out}'}\right) + F^{f_\mathrm{out}'} \tag{18}$$

$$F^{f_\mathrm{out}} = \mathrm{Conv}^{3\times3}\left(F^{f_\mathrm{out}''}\right) + F^{f_\mathrm{out}''}. \tag{19}$$

Among them, F^{f_out} represents the final fusion feature, while $F^{f_\mathrm{out}''}$ and $F^{f_\mathrm{out}'}$ are intermediate results generated during the calculation process of F^{f_out}. Specifically, $F^{f_\mathrm{out}'}$ is the preliminary fusion result derived from the double-residual structure.

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The dual-residual structure in each fusion branch adaptively adjusts the weighting and combination of different modal information, enabling the fused features to better represent the critical content of the two modalities. The 3×3 convolutional blocks with residual connections facilitate deeper fusion while maintaining effective gradient propagation, ensuring that the fusion module can continuously learn and optimize fusion strategies. Additionally, the 3×3 convolution captures local spatial features, further enhancing the detailed expression of the fused features.

D. Loss Function

The loss function for rotated object detection in this article consists of a classification loss, l_{cls}, which determines the object category, and a regression loss, l_{reg}, which constrains the object's bounding box. The regression loss, l_{reg}, is further composed of the IoU loss, l_{box}, and the shape loss, l_{dfl}

$$L_t = \alpha_1 L_{\text{cls}} + \alpha_2 L_{\text{box}} + \alpha_3 L_{\text{dfl}} \tag{20}$$

where α_1, α_2, and α_3 are balance parameters, and the values of α_1, α_2, and α_3, as well as the calculation formulas for l_{cls}, l_{box}, and l_{dfl}, are consistent with the baseline.

Among them, the calculation method of the classification loss L_{cls} (multilabel) of the model is

$$L_{\text{cls}} = -\sum_{c=1}^{C} g_c \cdot \log \sigma(p_c) + (1 - g_c) \cdot \log(1 - \sigma(p_c)) \tag{21}$$

where C is the total number of classes, p_c is the model's prediction for the cth class, g_c is the true label for the cth class (either 0 or 1, indicating whether the class is positive or not), and $\sigma(p_c)$ is the probability of the predicted class after applying the sigmoid function.

The definition of l_{box} is

$$L_{\text{box}} = 1 - \text{ProbIoU}(P_{\text{obb}}, G_{\text{obb}}) \tag{22}$$

where P_{obb} represents the object bounding box predicted by the model and G_{obb} denotes the ground-truth bounding box of the real object. The Probiou calculates the Hellinger distance between the predicted and true rotated bounding boxes, which is then used to compute the loss.

$$L_{\text{dfl}} = ((t_r - t) \cdot \text{CE}(p, t_l)) + ((t - t_l) \cdot \text{CE}(p, t_r)) \tag{23}$$

where t is the continuous object value. p is the predicted probability distribution. t_l is the nearest lower bound of t. t_r is the nearest upper bound of t. The weights $(t_r - t)$ and $(t - t_l)$ quantify how close t is to each bound, and the DFL loss is computed as the weighted sum of the cross-entropy losses $\text{CE}(p, t_l)$ and $\text{CE}(p, t_r)$.

IV. Experiments

This section provides a comprehensive analysis of the performance of our proposed ADMPF. We begin by describing the datasets used in the experiments, followed by a detailed explanation of the implementation process and the results of the ablation study. We then present a comparative evaluation of ADMPF against several state-of-the-art algorithms on different datasets, demonstrating the effectiveness and superiority of our approach. In addition, we compare the computational efficiency of ADMPF with other advanced algorithms and discuss the transferability of the proposed algorithm across different settings.

A. Datasets

DroneVehicle is a dataset consisting of infrared and visible (RGB) image pairs from a UAV viewpoint for ground object detection [20]. It covers a wide range of scenarios at different times of the day in different areas such as city roads, car parks, residential areas, and so on. DroneVehicle contains 17 990 infrared and visible image pairs for training, 1469 for validation, and 8980 for testing. These images cover four downward-looking shooting angles of 0°, 15°, 30°, and 45°. Additionally, the dataset has five categories of ground objects, that is, car, bus, truck, van, and freight car.

VEDAI is a multispectral aerial dataset for ground object detection [30]. The dataset holds more than 1200 image pairs, each with strictly aligned infrared and visible (RGB) modalities. It covers different challenging scenarios, such as light changes and occlusions. Besides, nine categories of ground objects are contained. The nine types of objects in the dataset are car, pickup, truck, camping vehicle, boat, tractor, van, and "other."

B. Implementation Details

The method proposed in this article is implemented using PyTorch and is trained and tested on a workstation equipped with an NVIDIA A6000 GPU. The algorithm uses Yolov8s [31] as the benchmark and employs a backbone network with a Siamese architecture to extract features from different modalities.

For the DroneVehicle dataset, the input image size is set to 640×640 pixels, and the model is trained for 100 epochs to achieve optimal results. For the VEDAI dataset, the input image size is increased to 1024×1024 pixels, with the same number of training epochs. The algorithm utilizes the stochastic gradient descent (SGD) optimization method, with a momentum of 0.937. The learning rate follows a cosine decay schedule, where the maximum learning rate is set to 0.001 and the minimum to 0.00001. Nonmaximum suppression (NMS) with an IoU threshold of 0.5 is applied to eliminate duplicate bounding boxes, ensuring accurate detection results.

C. Ablation Studies

1) Quantitative Analysis: In this section, we perform ablation studies to assess the effectiveness of the various components designed in our model. The ablation experiments are conducted on the test section of the DroneVehicle dataset. The baseline model used for comparison is a Siamese backbone, a one-stage detector that achieves multiscale feature fusion through a simple addition operation.

a) *GLSA:* GLSA is designed to integrate global and local context information within images, enhancing key features relevant to the object detection task and laying

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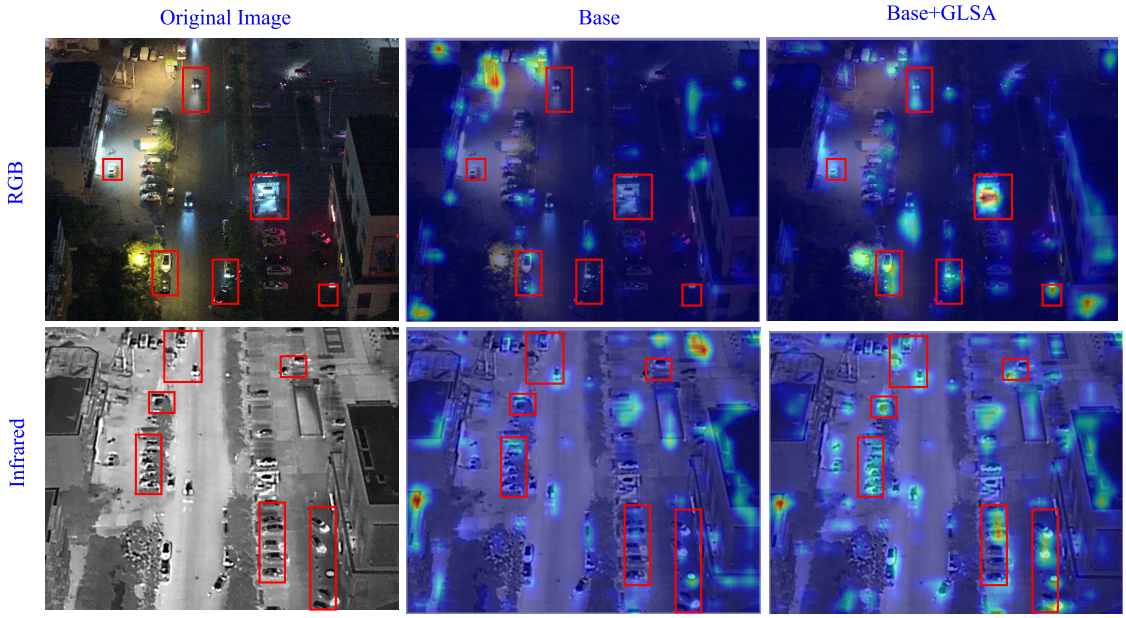


Fig. 4. Visualization of the MPFN fusion effect. "Base" depicts the model's attention area before applying the attention mechanism, while "Base + GLSA" shows the attention area after the application of the GLSA. The red box highlights the object distribution area where the model's attention is significantly enhanced following the attention's application.

TABLE I

RESULTS OF THE ABLATION EXPERIMENTS OF ADMPF WITH DIFFERENT COMPONENTS ON THE DRONEVEHICLE DATASET. ✓INDICATES THE ADDITION OF THE CORRESPONDING COMPONENT. NUMBERS IN RED INDICATE THE BEST PERFORMANCE, WHILE BOLD NUMBERS REPRESENT THE SECOND-BEST PERFORMANCE

Method	GLSA	MCIF	MPFN	mAP50	FLOPs
Baseline				74.94	42.22G
Baseline+GLSA	✓			76.73	**47.97G**
Baseline+GLSA+MCIF	✓	✓		**79.90**	47.97G
ADMPF(ours)	✓	✓	✓	82.05	68.80G

TABLE II

VERIFICATION OF ABLATION GAIN OF GLSA ATTENTION IN THE ADMPF METHOD. ✓INDICATES THE ADDITION OF THE CORRESPONDING COMPONENT. NUMBERS IN RED INDICATE THE BEST PERFORMANCE, WHILE BOLD NUMBERS REPRESENT THE SECOND-BEST PERFORMANCE

Method	CA	GLSA	mAP50
ADMPF*	✓		**81.01**
ADMPF(ours)		✓	82.05

the groundwork for filtering complementary information between different modalities. As shown in Table I, incorporating the GLSA module increases the model's mAP50 by 1.79%. This demonstrates that GLSA effectively enhances the key information in images by combining context information, resulting in improved detection performance. To further verify the role of the GLSA module, we compared the performance of the proposed GLSA attention mechanism with the CA [27] mechanism by replacing GLSA with CA in the ADMPF* method. The ADMPF method employs the GLSA attention specifically designed in this article. As shown in Table II, experimental results indicate that ADMPF outperforms ADMPF*, achieving a 1.04 increase in mAP50. This improvement is attributed to the GLSA attention's ability to incorporate both global and local contextual information, which plays a critical role in enhancing object detection in remote sensing images. Consequently, GLSA demonstrates superior performance over CA in this task.

b) *MCIF:* The MCIF focuses on filtering the crucial complementary information between different modal images, leveraging the GLSA. From the results in Table I, when both GLSA and MCIF are applied, the mAP50 of the model increases by 4.96% compared to the baseline and by 3.17% compared to the "Baseline + GLSA" model. This indicates that effectively filtering complementary information between different modalities significantly enhances the performance of multimodal fusion detection.

c) *MPFN:* The MPFN is designed to fully fuse the features enhanced by GLSA and the complementary information filtered by MCIF. Its goal is to merge the advantages of both infrared and RGB images to boost object detection performance. As shown in Table I, under the combined action of GLSA, MCIF, and MPFN, the mAP50 of the model reaches 82.05%, an improvement of 2.15% over the "Baseline + GLSA + MCIF" model. This demonstrates that MPFN efficiently integrates the strengths of infrared and RGB images, further improving detection accuracy.

Since the method in this article fuses features from three different scales of infrared and RGB images, GLSA was applied to the features at these three scales for both infrared and RGB images. According to experimental data, the application of GLSA leads to a slight increase in the algorithm's computational cost. However, since MCIF is built upon GLSA using

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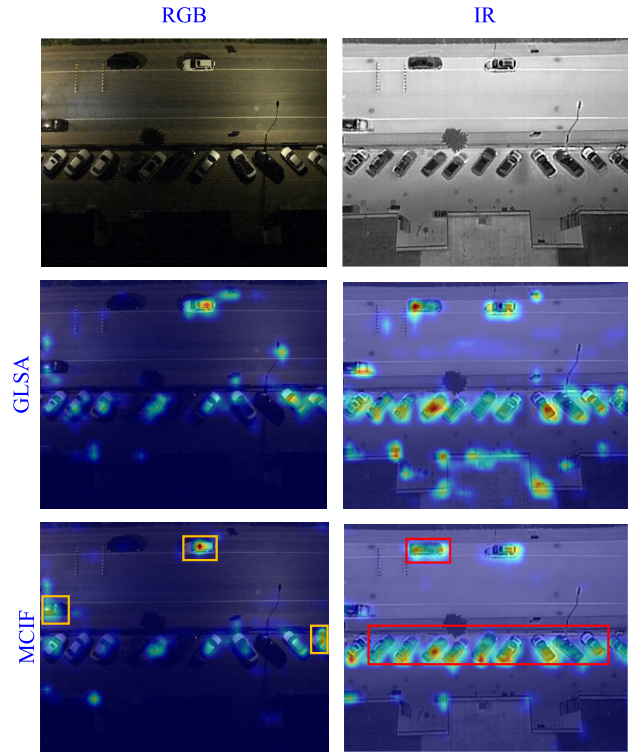


Fig. 5. Effect diagram of the MCIF. "MCIF" illustrates the distribution map of key complementary information after the application of the MCIF. The yellow box and the red box mark the detailed complementary information of the RGB image to the infrared image and the indicative complementary information of the infrared image to the RGB image.

the principle of conditional probability, the additional computational cost it introduces is minimal and can be ignored. Similar to GLSA, the increase in computational cost with the use of the MPFN module is based on a three-scale feature fusion process.

2) Qualitative Analysis: We conduct a visual analysis of image features to qualitatively verify the effectiveness of the modules designed in this article.

As depicted in Fig. 4, under complex lighting conditions in RGB images, the base features focus on a considerable amount of false alarm information, while their attention to actual object areas remains relatively low. However, after the application of the GLSA mechanism, the false alarms are significantly diminished, and the focus on the true object regions is notably enhanced.

In the case of infrared images, the sparse texture information results in a high rate of missed detections within the base features. The implementation of the GLSA mechanism markedly reduces the missed detection rate in the real object regions of infrared images. This indicates that the GLSA mechanism effectively integrates global and local context information, thereby substantially increasing attention to the actual object areas within the image. Consequently, it provides robust support for precise object detection.

For example, the texture of the vehicle under the light in the RGB image is clear, whereas the texture information of the vehicle is lacking in the infrared image, and thus the features in that region of the RGB image are enhanced. Identifying a

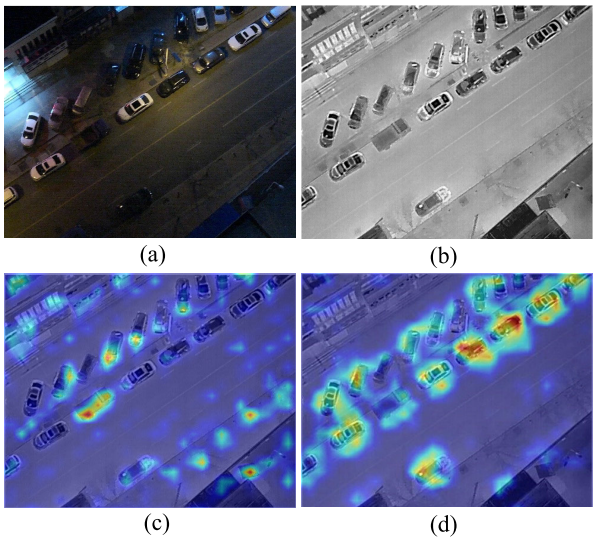


Fig. 6. Visualization of the MPFN fusion effect. "Baseline + GLSA + MCIF" indicates that the MPFN fusion structure is not utilized, while ADMPF denotes the use of the MPFN fusion structure. (a) RGB. (b) Infrared. (c) Baseline + GLSA + MCIF. (d) ADMPF.

vehicle in a dark location in the RGB image is challenging, whereas information about the outline of the vehicle at the corresponding location is apparent in the infrared image, and, therefore, features in that region of the infrared image are enhanced.

Fig. 5 illustrates the distribution of complementary information between different modalities as filtered by the complementary information filter proposed in this article. RGB images are abundant in fine-grained details but are sensitive to lighting conditions, making object detection challenging in low-light environments. As shown in Fig. 5, when lighting is adequate, RGB images can offer rich, fine-grained object information to supplement infrared images.

Conversely, infrared images are less affected by lighting changes but lack detailed texture information, making the detection of small objects in complex backgrounds difficult. Therefore, as depicted in Fig. 5, in low-light situations, infrared images can provide crucial cues about object positions to assist RGB images.

In summary, the MCIF designed in this study effectively integrates context information and extracts semantically rich complementary information between infrared and visible images, enhancing the overall object detection capability.

In Fig. 6, when compared with the "Baseline + GLSA + MCIF" without the adoption of MPFN, after using the MPFN fusion structure (ADMPF), it can more effectively focus on the distribution locations of the objects. Although the "Baseline + GLSA + MCIF" can also focus on some of the objects, there are numerous object areas that have not received due attention. This indicates that the MPFN fusion structure can better integrate the feature information of RGB–infrared images, enabling more accurate positioning.

D. Comparison Results With State-of-the-Art Methods on the DroneVehicle Dataset

In this section, we conduct a comprehensive evaluation of the proposed ADMPF method by comparing its performance

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TABLE III

COMPARISON OF ADMPF WITH STATE-OF-THE-ART METHODS ON THE DRONEVEHICLE DATASET. "RGB" AND "IR" REPRESENT THE DETECTION RESULTS OF VISIBLE IMAGES AND INFRARED IMAGES, RESPECTIVELY. "IR + RGB" MEANS THE RESULTS OF FUSION DETECTION OF INFRARED AND VISIBLE IMAGES. NUMBERS IN RED INDICATE THE BEST PERFORMANCE, WHILE BOLD NUMBERS REPRESENT THE SECOND-BEST PERFORMANCE

Detectors	Modal	car	bus	truck	freight car	van	mAP@0.5
RetinaNet [32]	RGB	67.50	62.05	28.24	13.72	19.26	38.10
R3DET [33]	RGB	79.30	76.00	42.20	24.50	28.50	50.10
Faster R-CNN [34]	RGB	67.88	66.98	38.59	26.31	23.2	44.59
RoITransformer [35]	RGB	68.13	70.55	44.17	29.08	27.64	47.91
ReDet [36]	RGB	69.48	77.37	47.87	31.46	29.03	51.04
Gliding Vertex [37]	RGB	75.77	68.05	46.08	33.75	38.72	52.48
Oriented R-CNN [38]	RGB	80.10	85.40	53.80	41.60	43.30	60.80
RetinaNet	IR	79.86	67.32	32.84	28.05	16.44	44.90
R3DET	IR	89.50	87.10	48.30	16.60	39.90	62.30
Faster R-CNN	IR	88.63	77.92	42.51	35.16	28.52	54.55
RoITransformer	IR	88.85	79.48	51.53	41.49	34.39	59.15
ReDet	IR	89.47	79.89	53.95	42.82	36.56	60.54
Gliding Vertex	IR	89.15	78.75	59.72	42.95	43.88	62.89
Oriented R-CNN	IR	89.80	89.30	57.40	53.10	45.40	67.00
UA-CMDet	IR+RGB	87.51	87.08	60.70	46.80	37.95	64.01
TSFADet [39]	IR+RGB	89.23	88.08	72.00	54.15	48.77	70.44
CIAN [40]	IR+RGB	90.10	89.10	63.80	60.70	50.30	70.80
AR-CNN [41]	IR+RGB	90.10	89.40	64.80	62.10	51.50	71.60
LF-MDet [42]	IR+RGB	82.20	86.60	73.60	59.60	57.00	71.80
C^2Former [21]	IR+RGB	90.20	89.80	68.30	64.40	58.50	74.20
GAGTDet [43]	IR+RGB	**90.82**	**90.46**	69.65	66.28	55.62	74.57
CLGNet [44]	IR+RGB	-	-	-	-	-	75.10
CALNet [45]	IR+RGB	90.30	89.11	76.15	62.97	58.46	75.39
SLBAF-Net [46]	IR+RGB	90.20	89.90	72.00	68.60	59.90	76.10
OAFA [47]	IR+RGB	90.30	90.30	**76.80**	73.30	66.00	**79.40**
ADMPF(ours)	IR+RGB	97.94	95.93	82.28	**69.34**	**64.77**	82.05

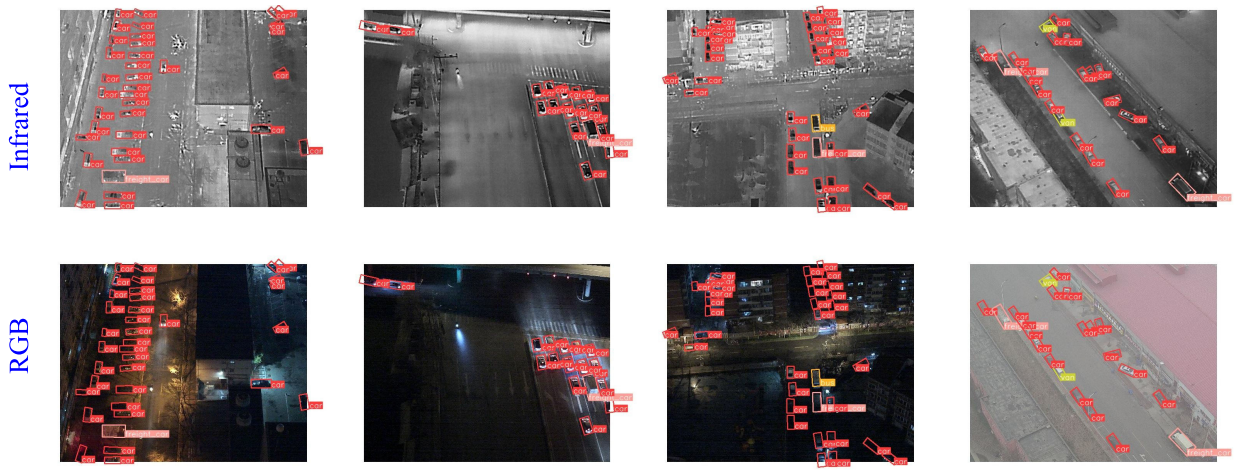


Fig. 7. Results of ADMPF fusion detection on the DroneVehicle dataset, shown on infrared and visible images, respectively.

against seven state-of-the-art single-modal aerial image object detection algorithms and 11 cutting-edge RGB–infrared aerial image object detection methods, all designed for oriented object detection. The evaluation is conducted on the widely used DroneVehicle dataset (Fig. 7).

As demonstrated in Table III, our ADMPF achieves an mAP50 score of 82.05% on the DroneVehicle dataset, surpassing all other compared methods. Specifically, ADMPF outperforms OAFA by 2.65%, SLBAF-Net by 5.95%, CALGNet by 6.66%, CLGNet by 6.95%, GAGTDet by 7.48%, C^2Former by 7.85%, LF-MDet by 10.25%, AR-CNN by 10.45%, CLAN by 11.25%, TSFADet by 11.61%, and UA-CMDet by 18.04%. Compared to the second-best OAFA method, the ADMPF method demonstrates distinct advantages by focusing on the precise handling of modality differences and improving the quality of fused features. This enables ADMPF to fully and efficiently leverage the complementary strengths of RGB and infrared images, resulting in superior performance in aerial object detection tasks when compared to OAFA. In terms of the five categories, ADMPF achieves AP50 scores of 97.94%, 95.93%, 82.28%, 69.34%, and 94.77% for car, bus, truck, freight_car, and van categories, respectively.

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TABLE IV

COMPARISON OF ADMPF WITH STATE-OF-THE-ART METHODS ON THE VEDAI DATASET. "RGB" AND "IR" REPRESENT THE DETECTION RESULTS OF VISIBLE IMAGES AND INFRARED IMAGES, RESPECTIVELY. "IR + RGB" MEANS THE RESULTS OF FUSION DETECTION OF INFRARED AND VISIBLE IMAGES. NUMBERS IN RED INDICATE THE BEST PERFORMANCE, WHILE BOLD NUMBERS REPRESENT THE SECOND-BEST PERFORMANCE

Detectors	Modal	Car	Pickup	Truck	Camping Vehicle	Boat	Tractor	Van	Other	mAP50
YOLOrs [48]	IR	82.03	73.92	54.21	63.80	21.97	54.39	43.38	43.99	54.71
YOLOFusion [49]	IR	86.70	75.90	77.1	66.60	70.70	62.30	**84.30**	43.0	70.8
SuperYOLO [19]	IR	87.90	81.39	61.56	76.90	46.08	60.56	71.00	39.39	65.60
YOLOrs	RGB	85.25	72.93	50.65	70.31	18.65	76.77	38.92	42.67	57.00
YOLOFusion	RGB	91.10	82.30	78.30	75.10	71.80	**81.20**	62.20	33.30	71.90
SuperYOLO	RGB	90.30	82.66	68.55	76.69	58.08	79.48	70.30	53.86	72.49
TPH-YOLO [50]	RGB	84.00	76.40	62.90	60.70	23.70	63.50	57.30	38.30	58.40
FFCA-YOLO [51]	RGB	89.60	85.70	85.70	78.70	61.50	81.80	67.00	48.60	74.80
YOLOrs	IR+RGB	84.15 7	78.27	52.60	68.81	21.47	67.88	57.91	46.75	59.73
YOLOFusion	IR+RGB	**91.70**	**85.90**	**78.10**	78.90	**71.70**	71.90	75.20	54.70	75.90
SuperYOLO	IR+RGB	91.13	85.66	70.18	**79.30**	60.24	80.41	76.50	57.33	75.09
ICAFuse [52]	IR+RGB	-	-	-	-	-	-	-	-	76.62
CLGNet [44]	IR+RGB	-	-	-	-	-	-	-	-	**80.20**
ADMPF(ours)	IR+RGB	94.45	90.25	74.41	84.74	85.46	74.74	91.86	**57.16**	81.63



Fig. 8. Results of ADMPF fusion detection on the VEDAI dataset, shown on infrared and visible images, respectively.

Notably, except for the freight_car and van category, ADMPF ranks highest for the other three categories when compared to other comparison methods. The data above clearly demonstrates that the ADMPF method significantly outperforms other comparison methods in terms of detection performance, effectively validating the effectiveness of the approach proposed in this article.

E. Comparison Results With State-of-the-Art Methods on the VEDAI Dataset

In this section, the proposed method ADMPF is compared with other state-of-the-art RGB–infrared aerial image object detection methods on the VEDAI dataset (Fig. 8).

As shown in Table IV, ADMPF outperforms the other compared methods with an mAP50 score of 81.63% on the VEDAI dataset. It is shown that all four comparative algorithms benefit from the fusion of infrared and RGB images, which can be reflected by the improved mAP50 values. However, the proposed ADMPF still achieves a superior performance. Among the RGB–infrared fusion detection algorithms involved in the comparison, CLGNet achieves the highest mAP50 score of 80.20%. In comparison, ADMPF improves the mAP50 by 1.43%. Compared to the second-best CLGNet method, the ADMPF method reduces the impact of redundant information between modalities on fused features. It achieves this by employing a specially designed MCIF and fusion network, effectively enhancing modality-specific features through a multibranch architecture. As a result, ADMPF demonstrates superior performance compared to CLGNet in evaluations. When compared with other fusion detection methods, ADMPF shows significant performance advantages in object detection across five categories: car, pickup, truck, camping, and boat. The "Other" label encompasses a wide range of object categories, with considerable variation in object properties within this group. Consequently, the detection performance of ADMPF for "Other" objects is relatively lower compared to

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TABLE V
COMPARISON OF COMPUTATIONAL EFFICIENCY AND PERFORMANCE METRICS FOR DIFFERENT DETECTION ALGORITHMS. NUMBERS IN RED INDICATE THE BEST PERFORMANCE, WHILE BOLD NUMBERS REPRESENT THE SECOND-BEST PERFORMANCE

Detectors	Params	FLOPs	FPS	mAP	Type
CIAN [40]	–	**70.36G**	21.7	70.23	One-stage
LF-MDet [42]	**38.7M**	77.7G	–	71.80	One-stage
AR-CNN [41]	–	104.3G	18.2	71.58	Two-stage
TSFADet [39]	104.7M	109.8G	18.6	73.06	Two-stage
CAGTDet [43]	–	120.6G	17.8	74.57	Two-stage
C2Former [21]	100.8M	89.9G	–	74.20	Two-stage
OAFA [47]	–	–	**33.1**	**79.40**	One-stage
ADMPF(ours)	31.8M	68.80G	39.8	82.05	One-stage

its detection performance in other specific object categories. Since the average object size in the DroneVehicle dataset is larger than that in the VEDAI dataset, our method achieves better performance on the DroneVehicle dataset as shown in Tables III and IV. This also indicates that there is still significant room for improvement in detecting smaller objects using the proposed method.

F. Efficiency Comparison With State-of-the-Art Methods

The computational efficiency and parameter complexity of the proposed ADMFP algorithm were thoroughly analyzed and compared with other state-of-the-art methods (see Table V). All algorithms were tested on the DroneVehicle Dataset with a batch size of 1 and an input size of 640 × 640 pixels. The results demonstrate that ADMFP achieves a parameter count of only 31.8 M, significantly lower than methods like LF-MDet. Moreover, its FLOPs are just 68.80G, substantially reducing computational complexity compared to two-stage detectors such as TSFADet and CAGTDet. In terms of speed, ADMFP achieves an FPS of 39.8, considerably outperforming most two-stage methods and fully meeting real-time detection requirements. Despite its lightweight and computationally efficient design, ADMFP attains an impressive mAP of 82.05, outperforming existing algorithms in detection accuracy. These findings confirm that ADMFP balances efficiency and performance, making it a promising solution for real-world object detection tasks.

G. Analysis of Transportability

To evaluate the transferability of the proposed algorithm, we tested the model trained on the DroneVehicle dataset using the VEDAI dataset. As the object categories of DroneVehicle and VEDAI datasets do not fully overlap, we conducted a statistical analysis on the intersection of their object categories. The results, presented in Fig. 9, demonstrate that the ADMPF model achieves good transferability for the "car" category, with an AP50 of 78.75. However, the transfer performance for "truck" and "van" is relatively poor, with AP50 values of 23.23 and 14.91, respectively. Consequently, the overall mean transfer performance, represented by mAP50, is 38.96,

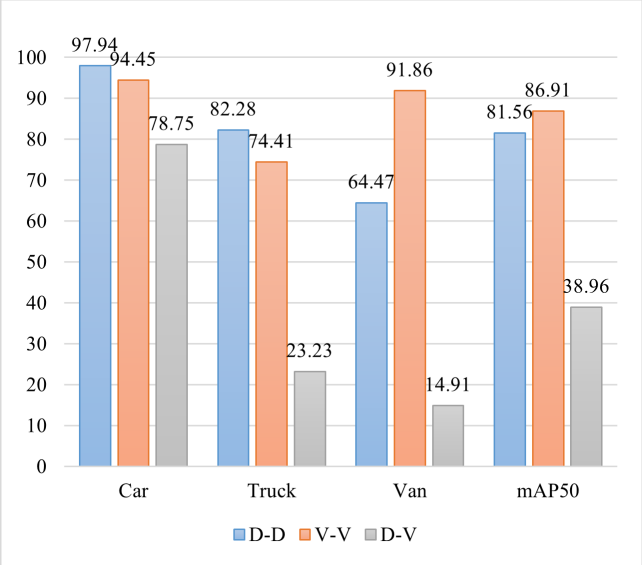


Fig. 9. Transferability Analysis of ADMPF (ours). "D-D": training and testing on DroneVehicle. "V-V": training and testing on VEDAI. "D-V": training on DroneVehicle and testing on VEDAI.

indicating a moderate level of transferability. However, it still exhibits a noticeable gap when compared to the model's performance within the same dataset (e.g., VEDAI to VEDAI).

We attribute these performance differences primarily to the following factors. First, the two datasets were captured in different geographical regions, likely resulting in significant intraclass variations for the same categories between DroneVehicle and VEDAI. The datasets exhibit distinct scene compositions, with DroneVehicle capturing predominantly urban environments and VEDAI centered on suburban settings. This divergence in scene characteristics likely introduced challenges for cross-dataset transferability, thereby contributing to the observed decline in performance.

V. CONCLUSION

In this work, we propose a multispectral aerial image object detection method, named ADMPF, based on RGB–infrared multibranch progressive fusion. The ADMPF method incorporates key modules such as GLSA, MCIF, and MPFN. The GLSA module integrates contextual information to enhance the key features related to the object detection task within single-modal image data, simultaneously providing the foundation for the MCIF module. Building upon GLSA, the MCIF module filters complementary information between infrared and RGB images efficiently and accurately, leveraging conditional probability theory. The MPFN module utilizes a multibranch structure to independently fuse the complementary and regular information from RGB and infrared images. It then performs a second-stage fusion of the previously fused complementary and regular information, thereby maximizing the complementary strengths of both infrared and RGB imagery. Extensive experimental results demonstrate that the proposed ADMPF method effectively merges the advantages of infrared and RGB imagery, yielding robust aerial object detection performance. Compared to state-of-the-art methods,

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our approach exhibits superior performance. In the future, we will focus on improving the detection capabilities of RGB–infrared fusion for small objects.

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